

EDUCATION

- **BTech, Electrical Engineering**, National Institute of Technology, Jaipur 2025

EXPERIENCE

ICICI Bank — Data Engineer, Mumbai *July 2025 – Present*

- Worked with IBM InfoSphere DataStage and Designer to schedule and monitor automated ETL jobs, ensuring reliable batch processing and timely data availability across enterprise systems.
- Analyzed and resolved UCIC (Unique Customer Identification Code) mismatches by investigating MDM deduplication rules and API matching logic, improving customer identity accuracy across downstream applications.
- Built and enhanced Azure ETL pipelines using Azure Data Factory and Azure Databricks to ingest and transform source-system data, delivering analytics-ready datasets to business and risk teams.
- Implemented PII and PCI data redaction using regular expressions to identify sensitive fields across on-premise Oracle and MySQL databases, strengthening data security and compliance controls.

PROJECTS

Multi-Agent Content Intelligence & Publishing System

- Designed a multi-agent system using LangGraph where specialised agents (research, drafting, quality-check) coordinate via shared state to autonomously generate and publish LinkedIn content.
- Exposed agent orchestration logic through a FastAPI microservice, enabling async job submission, streaming LLM responses, and webhook-based triggering from the n8n workflow layer.
- Containerised the full stack with Docker on GCP, managing session persistence, secret injection, and retry logic for production-grade reliability.

Conversational AI Assistant with Context Memory (WhatsApp)

- Built a production conversational AI assistant on WhatsApp using WAHA + LangChain, implementing session-based context management and a short-term memory buffer to maintain conversation continuity.
- Designed a vector-backed retrieval layer (ChromaDB) for FAQ grounding, reducing hallucinated responses on domain-specific queries.
- Deployed an async event-handling backend with FastAPI and Docker on a GCP Linux VM, integrating a rule-based AI classifier to filter unsafe or off-topic responses before delivery.

Agentic RAG Pipeline for Quantitative Alpha Research

- Built a RAG pipeline using LangGraph to orchestrate an autonomous research agent that ingests unstructured financial documents, retrieves relevant context via semantic search, and generates structured alpha signals using LLMs.
- Built a vector retrieval layer using ChromaDB with Gemini embeddings and cosine similarity search, grounding LLM outputs to reduce hallucination on domain-specific financial data.
- Implemented async Python workflows with queue-based parallel execution and SQL logging for experiment tracking; used the WorldQuant Brain API to programmatically submit alpha formulas and retrieve backtesting results.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **AI & ML:** PyTorch, TensorFlow, LSTM, NLP, LangChain, LangGraph, Hugging Face, ChromaDB
- **Database & Analytics Tools:** ETL (IBM DataStage), MDM, Azure Data Factory, Databricks
- **Data Visualisation:** Tableau, Power BI
- **Software:** Visual Studio Code, GitHub Desktop, MS Office (Excel, Word, PowerPoint), GCP
- **Others:** Dynatrace, Postman, Spark SQL, SQL Developer, n8n, Docker

LEADERSHIP

- **Executive, Think India MNIT Club:** Led the planning and execution of large-scale events with 5,000+ attendees, managing and coordinating a team of 20+ students to ensure smooth operations and delivery.
- **Founder, Unicorn Club (2022):** Built and scaled a community of entrepreneurs, organizing and hosting 30+ talks focused on practical business insights, collaboration, and knowledge sharing.